Universidad Rafael Landívar

Facultad de Ingeniería

Ingeniería en Informática y de Sistemas

Ciencia de Datos

Sección No. 02

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Fase 2

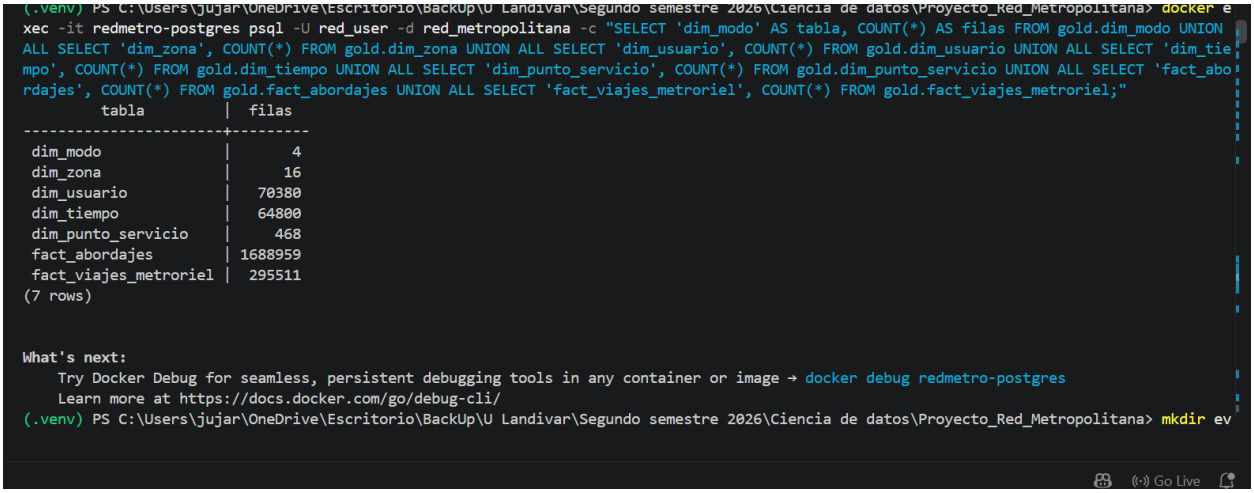
Julio Javier Robles Garcia - 1248823

Ana Lucia Chavez Penagos - 1067423

Guatemala, 12 de octubre de 2026

Desarrollo

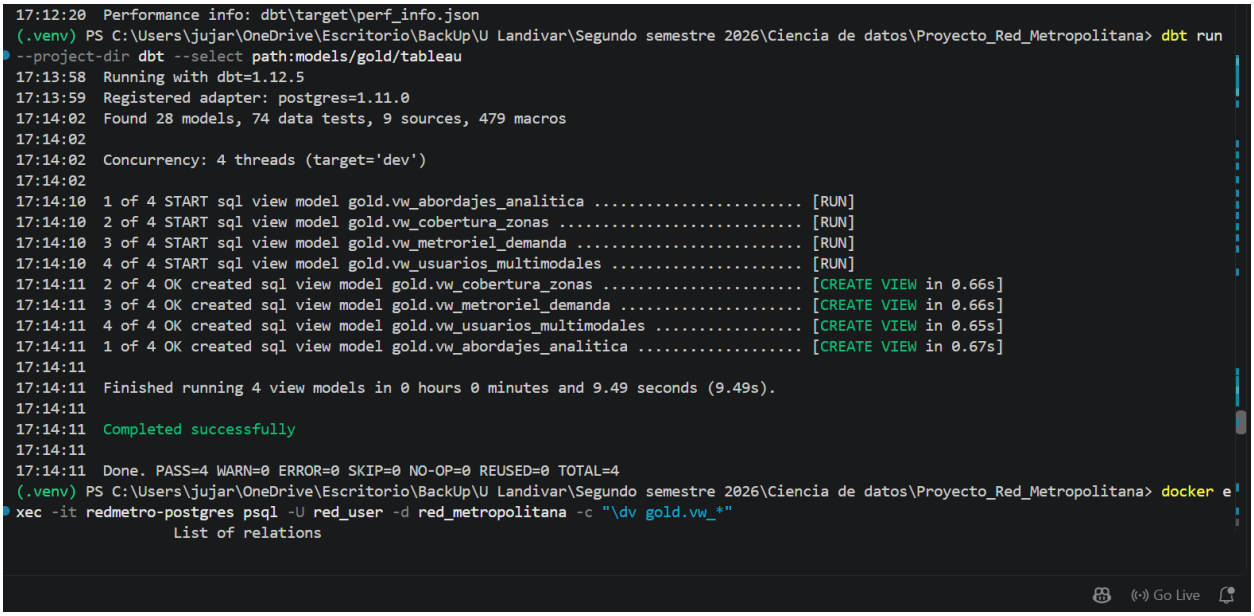
Captura 1 - Validación del entorno y disponibilidad de la capa Gold



```
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> docker e
xec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "SELECT 'dim_modo' AS tabla, COUNT(*) AS filas FROM gold.dim_modo UNION
ALL SELECT 'dim_zona', COUNT(*) FROM gold.dim_zona UNION ALL SELECT 'dim_usuario', COUNT(*) FROM gold.dim_usuario UNION ALL SELECT 'dim_tie
mpo', COUNT(*) FROM gold.dim_tiempo UNION ALL SELECT 'dim_punto_servicio', COUNT(*) FROM gold.dim_punto_servicio UNION ALL SELECT 'fact_abo
rdajes', COUNT(*) FROM gold.fact_abordajes UNION ALL SELECT 'fact_viajes_metroriel', COUNT(*) FROM gold.fact_viajes_metroriel;"
         tabla         |  filas
-----------------------+---------
 dim_modo              |       4
 dim_zona              |      16
 dim_usuario           |   70380
 dim_tiempo            |   64800
 dim_punto_servicio    |     468
 fact_abordajes        | 1688959
 fact_viajes_metroriel |  295511
(7 rows)

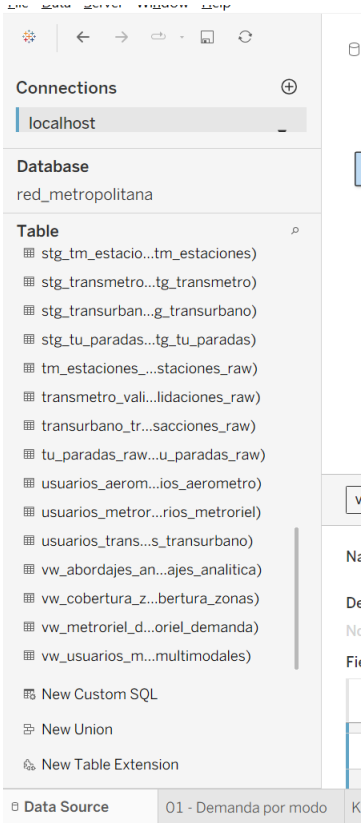

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug redmetro-postgres
    Learn more at https://docs.docker.com/go/debug-cli/
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> mkdir ev
```

Captura 2 – Creación de las vistas analíticas Gold para Tableau

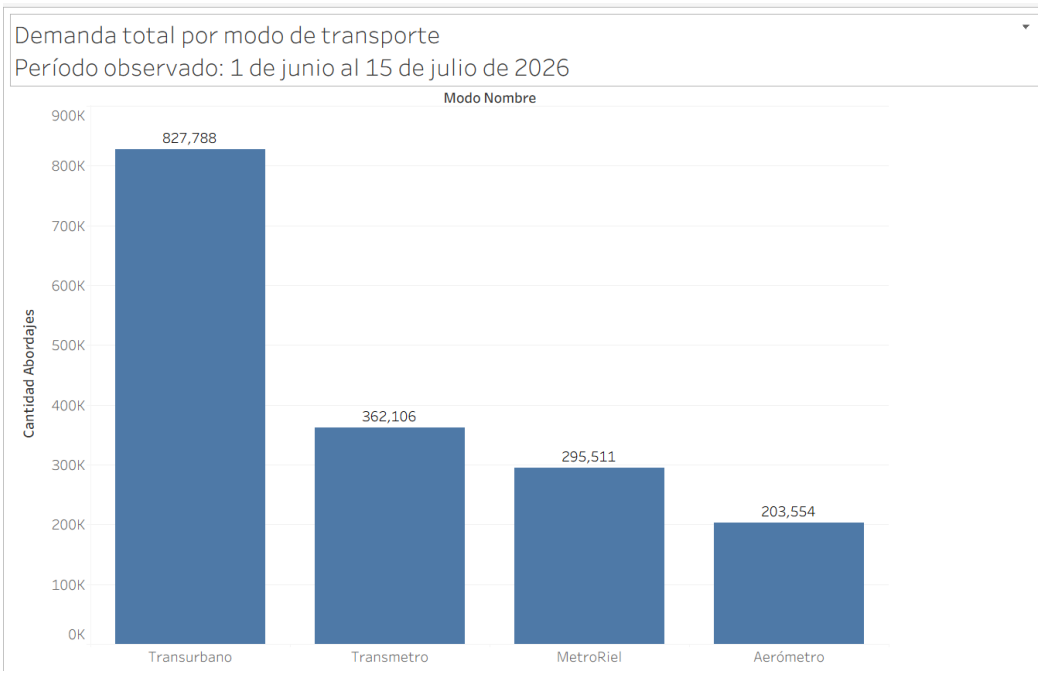


```
17:12:20  Performance info: dbt\target\perf_info.json
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> dbt run
--project-dir dbt --select path:models/gold/tableau
17:13:58  Running with dbt=1.12.5
17:13:59  Registered adapter: postgres=1.11.0
17:14:02  Found 28 models, 74 data tests, 9 sources, 479 macros
17:14:02
17:14:02  Concurrency: 4 threads (target='dev')
17:14:02
17:14:10  1 of 4 START sql view model gold.vw_abordajes_analitica ....................... [RUN]
17:14:10  2 of 4 START sql view model gold.vw_cobertura_zonas ........................... [RUN]
17:14:10  3 of 4 START sql view model gold.vw_metroriel_demanda ......................... [RUN]
17:14:10  4 of 4 START sql view model gold.vw_usuarios_multimodales ..................... [RUN]
17:14:11  2 of 4 OK created sql view model gold.vw_cobertura_zonas ...................... [CREATE VIEW in 0.66s]
17:14:11  3 of 4 OK created sql view model gold.vw_metroriel_demanda .................... [CREATE VIEW in 0.66s]
17:14:11  4 of 4 OK created sql view model gold.vw_usuarios_multimodales ................. [CREATE VIEW in 0.65s]
17:14:11  1 of 4 OK created sql view model gold.vw_abordajes_analitica ................... [CREATE VIEW in 0.67s]
17:14:11
17:14:11  Finished running 4 view models in 0 hours 0 minutes and 9.49 seconds (9.49s).
17:14:11
17:14:11  Completed successfully
17:14:11
17:14:11  Done. PASS=4 WARN=0 ERROR=0 SKIP=0 NO-OP=0 REUSED=0 TOTAL=4
(.venv) PS C:\Users\jujar\OneDrive\Escritorio\BackUp\U Landivar\Segundo semestre 2026\Ciencia de datos\Proyecto_Red_Metropolitana> docker e
xec -it redmetro-postgres psql -U red_user -d red_metropolitana -c "\dv gold.vw_*"
                List of relations
```

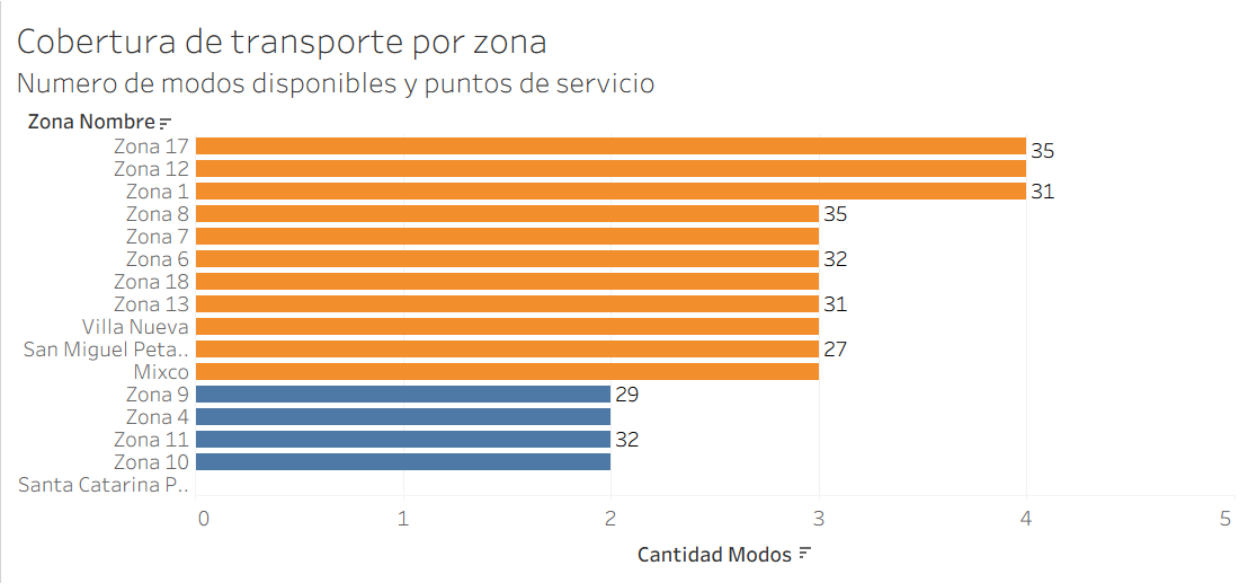
Captura 3 – Conexión de Tableau Desktop a la capa Gold



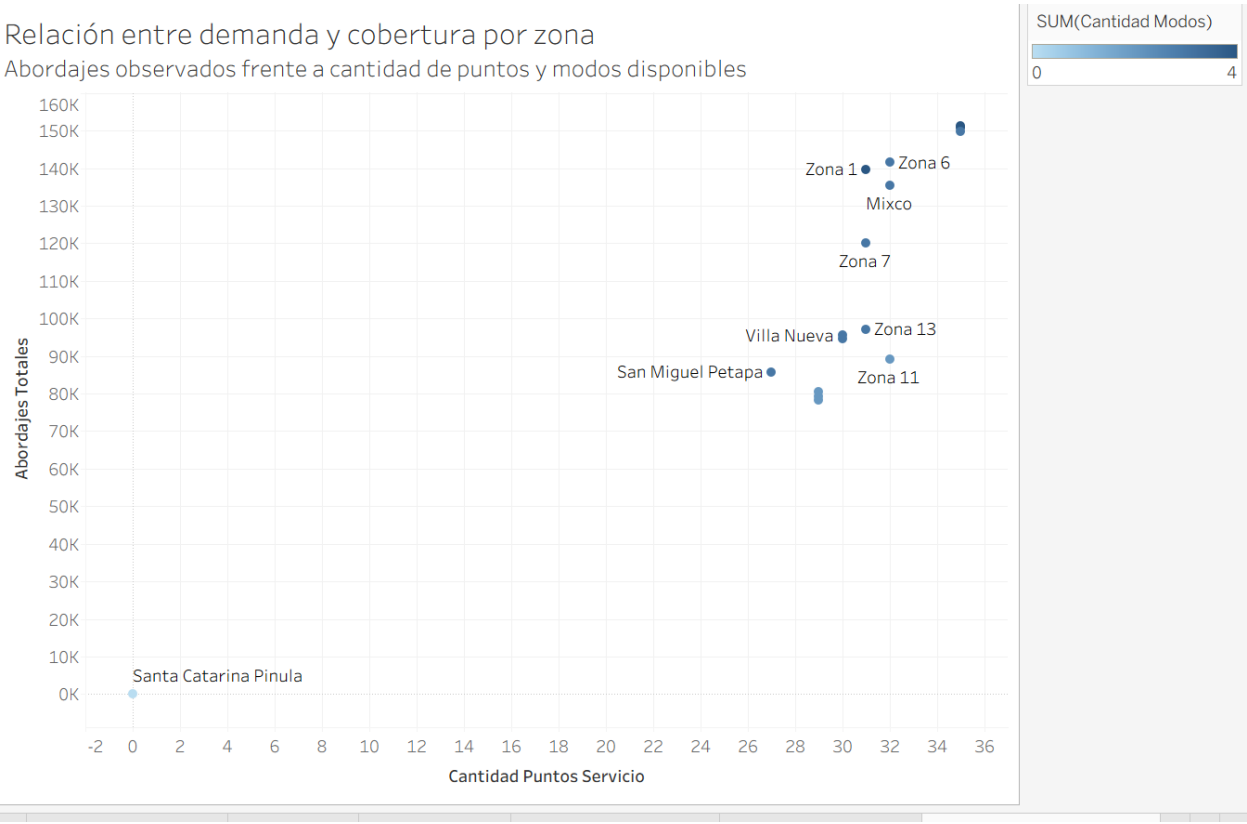
Captura 4 – Demanda total por modo de transporte



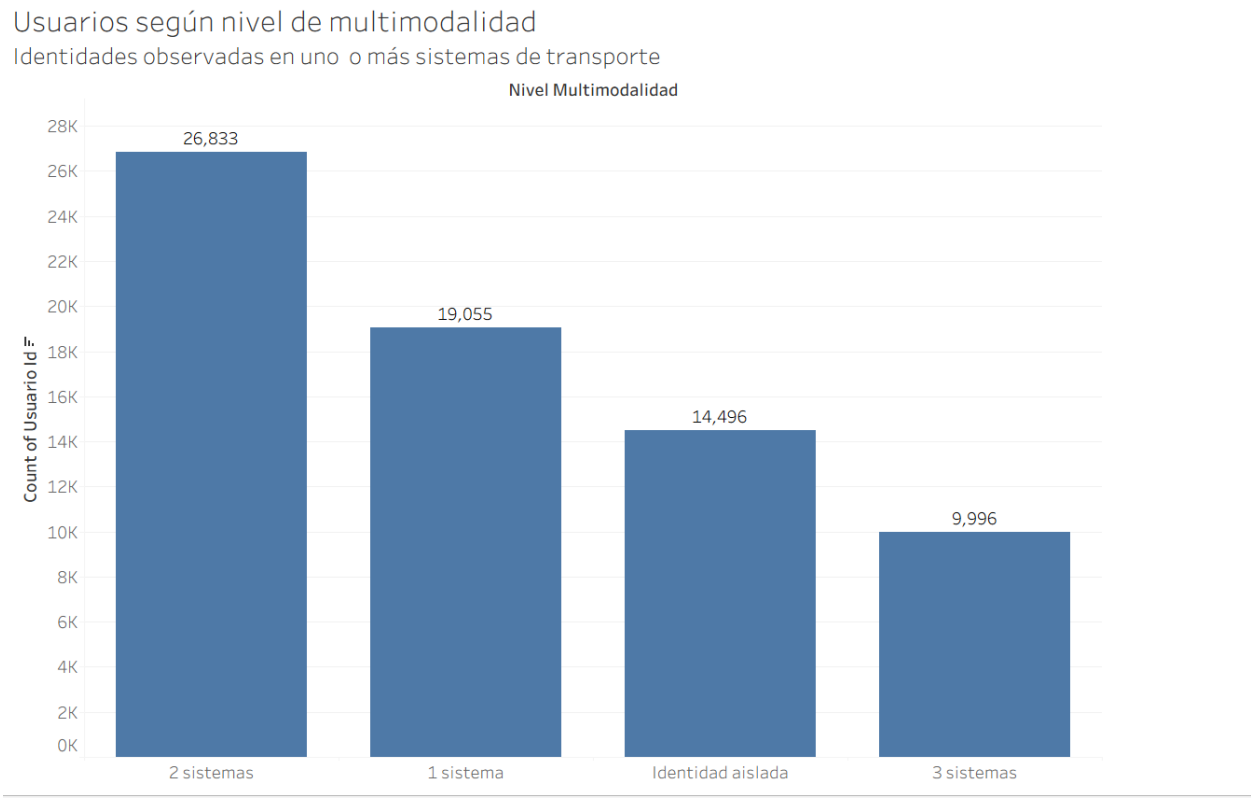
Captura 5 – Cobertura de transporte por zona



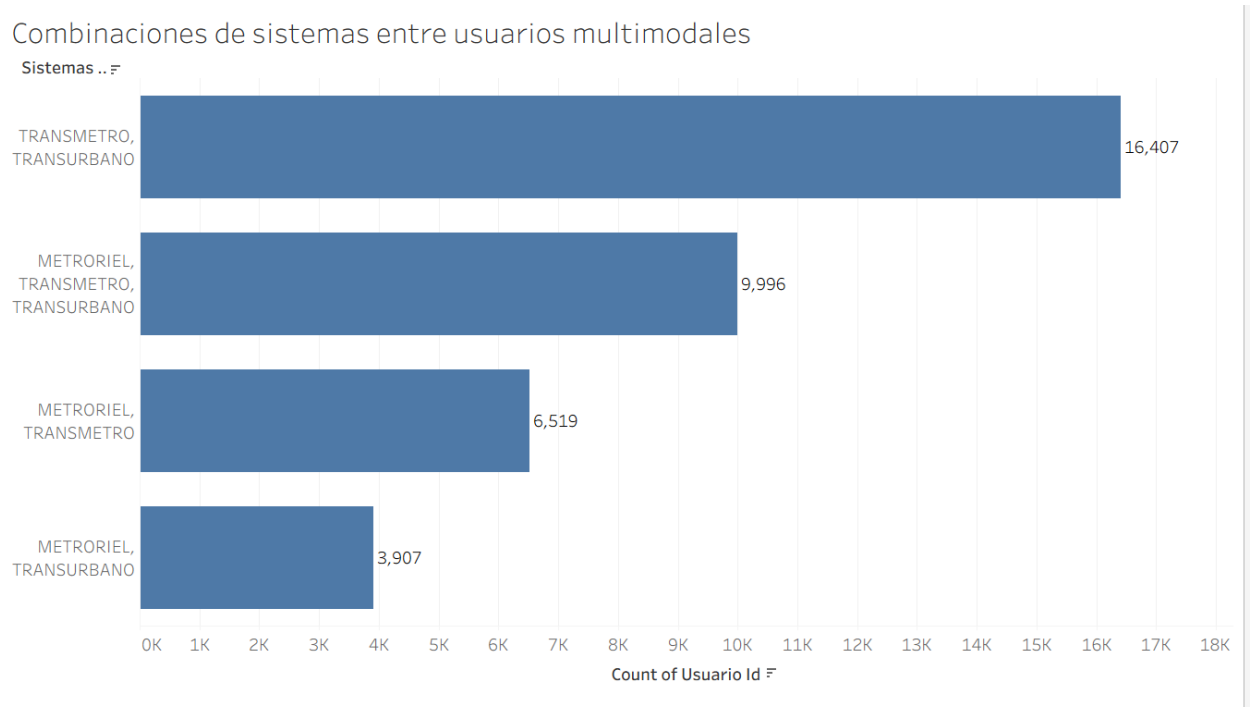
Captura 6 – Relación entre demanda y cobertura por zona



Captura 7 – Distribución de usuarios según nivel de multimodalidad



Captura 8 – Combinaciones de sistemas entre usuarios multimodales



Captura 9 – Validación de métricas del tablero contra la capa Gold

```
ent sql\queries\validacion_tableau_fase2.sql -Raw |
 modo_nombre | abordajes
-------------+-----------
 Transurbano |    827788
 Transmetro  |    362106
 MetroRiel   |    295511
 Aerómetro   |    203554
(4 rows)

 abordajes_totales
-------------------
           1688959
(1 row)

    tipo_horario    | abordajes | porcentaje
--------------------+-----------+------------
 Fuera de hora pico |    849926 |      50.32
 Hora pico          |    839033 |      49.68
(2 rows)
```

```
     zona_nombre      | cantidad_puntos_servicio | cantidad_modos | nivel_cobertura | abordajes_totales
----------------------+--------------------------+----------------+-----------------+-------------------
 Zona 17              |                       35 |              4 | 3 o más modos   |            151363
 Zona 12              |                       35 |              4 | 3 o más modos   |            150762
 Zona 8               |                       35 |              3 | 3 o más modos   |            149846
 Zona 6               |                       32 |              3 | 3 o más modos   |            141546
 Zona 1               |                       31 |              4 | 3 o más modos   |            139665
 Mixco                |                       32 |              3 | 3 o más modos   |            135451
 Zona 7               |                       31 |              3 | 3 o más modos   |            120148
 Zona 13              |                       31 |              3 | 3 o más modos   |             97103
 Villa Nueva          |                       30 |              3 | 3 o más modos   |             95585
 Zona 18              |                       30 |              3 | 3 o más modos   |             94626
 Zona 11              |                       32 |              2 | 2 modos         |             89052
 San Miguel Petapa    |                       27 |              3 | 3 o más modos   |             85789
 Zona 4               |                       29 |              2 | 2 modos         |             80576
 Zona 9               |                       29 |              2 | 2 modos         |             79255
 Zona 10              |                       29 |              2 | 2 modos         |             78192
 Santa Catarina Pinula |                        0 |              0 | Sin servicio    |                 0
(16 rows)

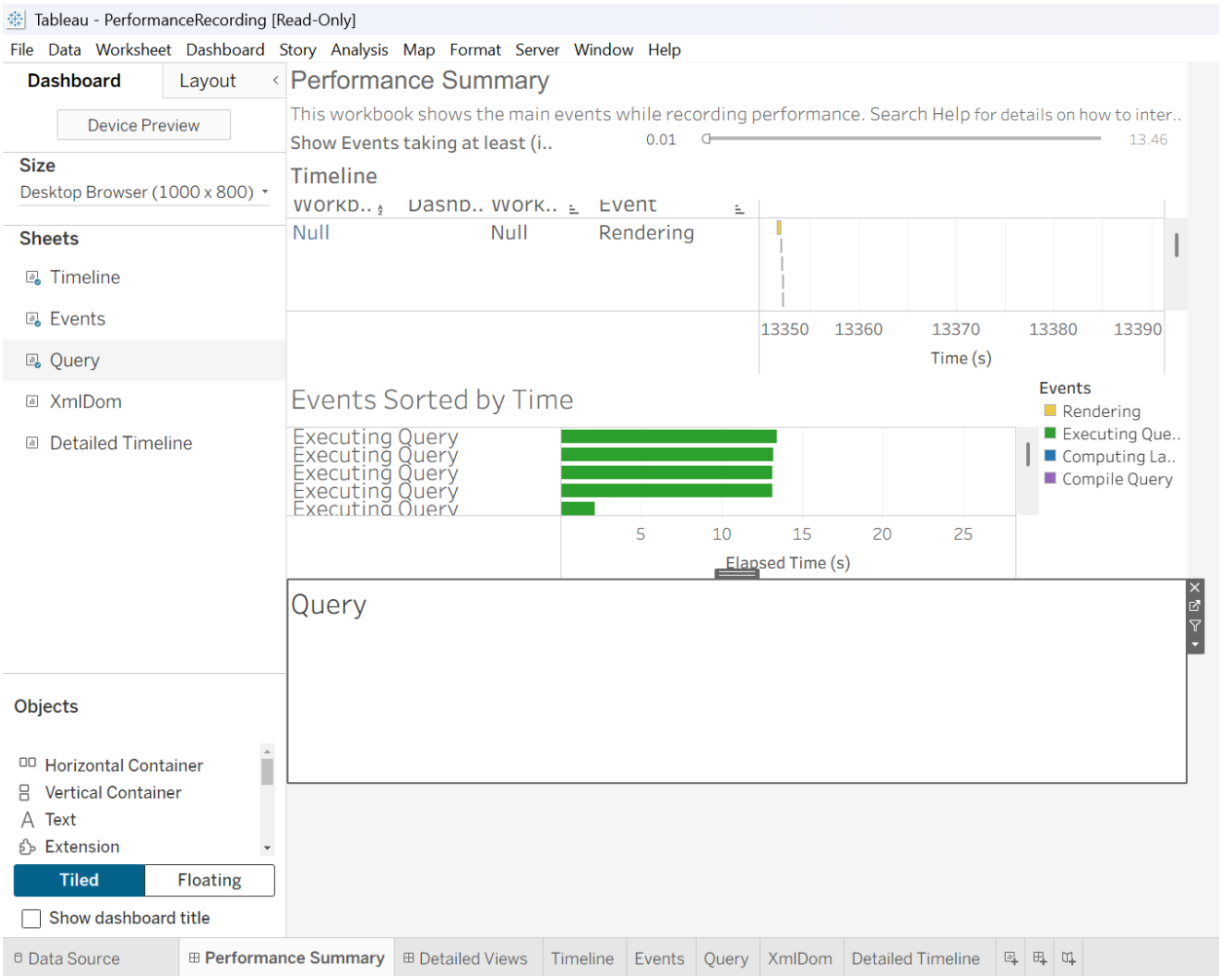
 zonas_sin_servicio
--------------------
                  1
(1 row)
```

```
 nivel_multimodalidad | usuarios
----------------------+----------
 2 sistemas           |    26833
 1 sistema            |    19055
 Identidad aislada    |    14496
 3 sistemas           |     9996
(4 rows)

 usuarios_multimodales | usuarios_totales | porcentaje_multimodal
-----------------------+------------------+-----------------------
                 36829 |            70380 |                 52.33
(1 row)

     tipo_zona       | abordajes | porcentaje
---------------------+-----------+------------
 Corredor MetroRiel  |    733182 |      43.41
 Fuera del corredor  |    955777 |      56.59
(2 rows)
```

Captura 10 – Registro de rendimiento del tablero Tableau


Tableau - PerformanceRecording [Read-Only]
File Data Worksheet Dashboard Story Analysis Map Format Server Window Help
Dashboard
Layout
Device Preview
Size
Desktop Browser (1000 x 800)
Sheets
Timeline
Events
Query
XmlDom
Detailed Timeline
Objects
Horizontal Container
Vertical Container
Text
Extension
Tiled
Floating
Show dashboard title
Performance Summary
This workbook shows the main events while recording performance. Search Help for details on how to inter..
Show Events taking at least (i..
0.01
13.46
Timeline
Null
Null
Rendering
13350
13360
13370
13380
13390
Time (s)
Events Sorted by Time
Executing Query
Executing Query
Executing Query
Executing Query
Executing Query
5
10
15
20
25
Elapsed Time (s)
Events
Rendering
Executing Que..
Computing La..
Compile Query
Query
Data Source
Performance Summary
Detailed Views
Timeline
Events
Query
XmlDom
Detailed Timeline